GenAI's Four Crises

A Two-Tier Master Synthesis of 23 Audited Papers

Audit synthesis – HEC Paris

Tier 1: Executive Overview | Tier 2: Five Deep Dives

Trust / UI ▪ Economic Displacement ▪ Data Consent ▪ Macro-Productivity

Tier 1

The Grand Synthesis

Four streams. Twenty-three papers. One feedback loop.

Twenty-three audited papers cluster into four crises

Stream	Papers (audited 2026-04)	Core question
Trust / UI	Spatharioti 25, Li & Aral 25, Sharma 24, Wardle 25, Liu 23, Jazwinska 24, Jazwinska 25, Potthast 20	Does the interface tell users what to verify?
Economic	Eloundou 24, Noy 23, Demirci 25, Goldberg 26, Agrawal 25, Humlum 25, Burtch 24, Liu (Labor) 25, Lyu 25	Where is GenAI taking work, and from whom?
Data consent	Cooper 25, Yang 24, Fletcher 24, Kim 25, Longpre 25	Will the training corpus survive the technology built on it?
Macro-prod.	Acemoglu 25, Aggarwal (GEO) 24, Khosravi 26, Larsen 25, Liang 25	What is the speed limit on GenAI's contribution to growth?

Each Tier 2 deep dive draws from one cell. The four cells share one finding: the constraints bind on each other.

Audit log (`claudework/AUDIT_LOG.md`); five synthesis decks D1–D5.

Lab efficiency does not survive contact with the macro envelope

Layer	What the data says	Reconciliation
Lab gain (Noy & Zhang 23)	Time -0.8 SD (TOT -16.51 min); grade $+0.4$ **SD**; 58/173 submitted unedited	Drafting halves; editing doubles. Substitution at the draft step, not ideation.
Macro envelope (Acemoglu 25)	4.6% of GDP $\times\ \bar{\pi}\ =\ 14.4\% \Rightarrow$ $d\ln\text{TFP} =\ 0.66\%$, 10-yr GDP $\leq$ 1.56%	Goldman 7%, McKinsey 1.5–3.4 pp/yr cannot be reconciled with task-level estimates inside Hulten.
Where they meet (Demirci 25; Goldberg 26)	Writing freelance posts -21%; non-GenAI artists -29% on Adobe Stock; surviving budgets *up*	Narrow, sharp dislocations *exactly as Hulten predicts* when 4.6% of GDP is exposed at 14%.

Both numbers are right. 80% of workers exposed (Eloundou 24) and 1.56% GDP (Acemoglu 25) describe the *same* economy: a wide-but-shallow shock concentrated where AI's curvature $\eta_{AI} \leq 1/q^2$ holds (Agrawal, Gans & Goldfarb 25, Prop. 3).

See Tier 2 Deep Dive 4 (Productivity Gap) for the full chain.

Direct answers train on a web that is closing

#	Mechanism	Empirical anchor (paper, number)
1	Direct-answer interface	Potthast 20: four risks – misattribution, source erosion, market distortion, loop pollution
2	Citation integrity falls	Liu 23: precision 74.5%, recall 51.5%; $r = -0.96$ fluency × citation precision
3	Misattribution scales	Jazwinska 24/25: 60% wrong across 8 chatbots; Grok 3 94%; premium tiers more confidently wrong
4	Pageviews collapse	Khosravi 26: Wikipedia -15%; $35.7M–$102.1M annual donations at risk
5	Premium supply with-draws	Fletcher 24: 48% of top news blocked OpenAI by end-2023; legacy print 57%; none reversed
6	Consent withdrawn	Longpre 25: 25.9% of OpenAI HEAD_C4 tokens restricted via robots.txt in 12 mo (vs. 1.0% Googlebot)
7	Joint policy required	Yang & Zhang 24, Prop. 10–12: in scarce data, generous fair use HARMS AI development

The loop closes on itself. Each turn cheapens the next – until the legal backstop (Cooper 25: $1.5B Bartz v. Anthropic) and the consent backstop both bind.

See Tier 2 Deep Dives 2 (Direct Answer Dilemma) and 3 (Closing Web).

The four crises are joint constraints, not independent levers

Deep dive	**One-line claim**	**Where it binds the others**
1. Veneer of Search	Interface earned the trust; the model did not (Spatharioti 25: 47% accuracy at 4.41/5).	If users do not verify, displaced labor (DD 4) cannot route around errors.
2. Direct Answer Dilemma	Direct answers are an externality on the web that produced them.	Erodes the corpus that DD 3 measures closing.
3. Closing Web	Running out of consent, not data (Longpre 25: 25.9%).	Fixes Yang & Zhang's data-scarce regime as the binding policy frame.
4. Productivity Gap	The macro is small (1.56%); dislocations are not (-21% writing).	Caps the upside that justifies the consent and trust costs above.
5. Joint Copyright	f and ϕ are economically linked (Yang 24, Prop. 12).	The policy lever that lets DD 3 and DD 4 be solved together.

The synthesis claim: no single deep dive's reform works alone. Restoring user judgement (DD 1), repricing direct answers (DD 2), reopening the data commons (DD 3), and lifting the macro ceiling (DD 4) are tied through one policy surface (DD 5).

Tier 2

Five Deep Dives

The full audit, one stream at a time.

Deep Dive 1

The Veneer of Search

How AI Interfaces Earn Trust They Haven't Earned

Spathariotti et al. (2025), *CHI '25* | Li & Aral (2025), *arXiv:2504.06435*
Sharma, Liao & Xiao (2024), *CHI '24* | Wardle, Urbani & Wang (2025), *JMIR* 27, e79961

1.6 min **47%** **4.41/5**

AI search cut task time in half.
It halved accuracy on a hard task.
Users rated the experience *best of all*.

Spatharioti et al. (2025), Exp. 1, $N = 90$ MTurk; vs. traditional search 3.4 min, 93% accuracy, 3.10/5.

AI search wears four veneers — each builds unearned trust

Veneer	What it looks like	What it does	Source
Speed	"AI search is fast"	Hides errors users cannot catch	Spatharioti
References	"AI cites its sources"	Builds trust even when fake	Li & Aral
Search	"User issues queries"	Most queries are confirmatory	Sharma
Utility	"AI is useful"	Overrides users' own distrust	Wardle

Veneer 1: speed that hides errors

Faster searches, worse decisions, higher satisfaction.

LLM search halved task time — and halved accuracy on the hard task

Outcome	Traditional search	LLM-based search
Median task duration	3.4 min	1.6 min ($p < .001$)
Easy-task accuracy	92.3%	95.3% (NS)
Hard-task accuracy	**93%**	**47%** ($p < .001$)
Number of queries	2.5	1.9 ($p = .02$)
Query complexity (1–5)	1.8	3.4 ($p < .001$)

On the hard task, 30/51 LLM users issued only *one* query; none re-queried after a wrong answer.

Spathariotí et al. (2025), Exp. 1, Figs. 2–3; 4Runner adversarial item.

Users prefer LLM search 4.41 vs. 3.10 — and cannot detect its errors

Self-reported measure	Traditional	LLM
Overall search experience (1–5)	3.10	**4.41** ($p < .001$)
Perceived reliability (1–5)	∼4	∼4 (NS, $p = .91$)

The interface fix exists — but only if it is added

Confidence highlighting (red on low-probability tokens, Exp. 2) more than doubled hard-task accuracy from 26% to 58% ($p < .01$). The fix lives in the UI, not the model.

Spatharioti et al. (2025), Exp. 1 & 2, Figs. 4, 6.

Veneer 2: trust from interface, not content

The references work *whether or not they are real*.

Reference links increase trust — and hallucinated ones work just as well

Treatment	β on trust	β on share
References (vs. plain GenAI)	**+0.091** ***	**+0.121** ***
Hallucinated vs. valid references	-0.021 (NS)	-0.008 (NS)
Uncertainty highlighting	-0.157^{***}	-0.179^{***}
Negative social feedback	-0.213^{***}	-0.286^{***}
GenAI label vs. traditional (same content)	-0.043^{*}	-0.141^{***}

The mere *presence* of citations creates trust, regardless of whether they resolve. Most vulnerable: lower education, less tech experience, infrequent GenAI users.

Li & Aral (2025), pre-registered RCT, $N = 4{,}927$ Prolific, Figs. 3–4.

Veneer 3: confirmatory querying disguised as search

When the interface answers, the user stops looking.

Conversational search induces 10× more confirmatory querying than web search

Measure	WebSearch	ConvSearch(Ref)
Confirmatory query rate	1.46%	15.0–16.2%
Confirmatory agreement (post-task)	0.80	1.79–1.89
Confirmatory trust	0.35	0.79–0.89

ANCOVA $F(2,100)=4.71$, $p=.01$ on querying; $F=6.61$, $p=.002$ on agreement.
Cohen's D up to 0.78.
Inline references did not change this: ConvSearchRef behaved like ConvSearch.

Sharma, Liao & Xiao (2024), Study 1, $N=115$ Prolific, Table 1.

Consonant LLMs polarize — but dissonant LLMs do not correct it

Measure	Consonant	Neutral	Dissonant
Confirmatory query rate	42.9%	15.6%	12.3%
Confirmatory arguments in essays	51.7%	34.8%	15.6%
Confirmatory agreement	2.44	1.84	1.51

The asymmetry is the result. Consonant LLMs amplify polarization with Cohen's D up to 1.19; dissonant LLMs fail to depolarize on every measure (all $p > .05$ vs. Neutral). Designing "opposing-view" tools will not undo selective exposure.

Sharma, Liao & Xiao (2024), Study 2, $N = 223$, Table 3.

Veneer 4: utility that overrides distrust

Users adopt what they doubt — if it is convenient.

56% of users adopt AI for health information *despite* distrusting it

- **15/27 (56%)** used AI-generated health info despite expressing skepticism, when it "helped power further investigation."
- **14/27 (52%)** already integrate ChatGPT into their health-search routines; **23/27 (85%)** reported favorably on its formatting or efficiency.
- AI is used as a "starting point": keyword generation, summarization, then verified on Google or with a doctor — *utility*, not credibility, drives integration.
- Only 5/27 (19%) understood that training data shapes LLM outputs.
- **2/3** expressed concern that ChatGPT does not cite sources — yet still used it.

Wardle, Urbani & Wang (2025), think-aloud protocol, $N = 27$, JMIR 27:e79961.

A skeptic would say these are lab effects, not real-world behavior

The objection. Spatharioti uses MTurk ($N = 90$, no demographics); Sharma uses Prolific ($N = 115$–223, closed-world docs); Wardle is qualitative, $N = 27$ from four East Coast cities. Only Li & Aral has scale, and even that is a controlled vignette.

The response.

- The effect sizes are large. Cohen's $D > 1$ in Sharma; a 46-point accuracy gap in Spatharioti; β's significant at $p < .001$ in Li & Aral. Small samples find them anyway.
- Four very different methods converge. Between-subjects RCT, factorial RCT, large pre-registered RCT, qualitative think-aloud — all four point to interface-driven effects exceeding content-driven ones.
- The lab is the world. Li & Aral's exposure audit: **42%** of US Google searches now return AI Overviews; **56%** for general-knowledge queries. The vignettes describe what most users now see.

Restoring user judgment requires interface-level, not model-level, fixes

Veneer	Reform	Concrete evidence
Speed	Confidence highlighting on uncertain tokens	Spatharioti Exp. 2: 26%→58% accuracy
References	Verified-source badges; degrade UI on broken links	Li & Aral: refs trusted regardless of validity
Search	Default to neutral retrieval; flag opinion drift	Sharma: dissonant LLM does *not* fix it
Utility	AI literacy on training data; source labels	Wardle: only 19% understood data shapes outputs

Each fix is interface-side; none requires changing the underlying model.

The interface earned the trust.

The model did not.

Deep Dive 2

The Direct Answer Dilemma

How AI Search Closes the Attribution Loop — and Who Pays for It

Potthast et al. (2020), *SIGIR Forum* | Liu, Zhang & Liang (2023), *arXiv:2304.09848*
Jazwinska & Chandrasekar (2024, 2025), *Tow Center / CJR* | Aggarwal et al. (2024), *KDD '24*
Khosravi et al. (2026), *arXiv preprint* | Larsen et al. (2025) | Liang et al. (2025), *Patterns*

60%

of citation queries answered *wrong* by eight AI search engines.

Grok 3: 94% wrong. Most wrong answers came with no hedging.

The interface earned the user's trust. The citations did not.

Jazwinska & Chandrasekar (2025), *Tow Center / CJR*; 1,600 queries across 8 chatbots.

Potthast (2020) named four risks of the direct-answer interface — all four now have data

Potthast risk (2020)	Mechanism	Empirical confirmation
Misattribution	Answer drops the source	Liu 23; Jazwinska 24/25
Source erosion	Click never happens	Khosravi 2026
Market distortion	Ad auction reshapes	Larsen 2025
Loop pollution	Synthetic text re-enters web	Aggarwal 24; Liang 25

Potthast, Hagen & Stein (2020), *SIGIR Forum* 54(1):14.

Citation is broken at the source

Direct-answer interfaces grade their own homework.

The better the answer reads, the worse it cites — r $= -0.96$

Engine	Citation recall	Citation precision
Bing Chat	58.7%	89.5%
NeevaAI	67.6%	72.0%
perplexity.ai	68.7%	72.7%
YouChat	11.1%	63.6%
Mean (4 engines)	**51.5%**	**74.5%**

Across 1,450 statements: Pearson $r = -0.96$ between citation precision and the perceived fluency/utility of the answer. Better-sounding answers cite worse.

Liu, Zhang & Liang (2023), Tables 3–4; human eval over four generative search engines, May 2023.

ChatGPT Search misattributed 153 of 200 publisher quotes — and declined only 7 times

Response category	Count / 200	Share
Correct (publisher + date + URL)	47	23.5%
Partially correct	57	28.5%
Wrong	89	44.5%
Declined / unsure	7	3.5%

Even publishers *with* OpenAI licensing deals were misattributed. NYT (in active lawsuit, blocked) was cited via plagiarized copies.

Jazwinska & Chandrasekar (2024), *Tow Center / CJR*, Nov. 2024; 200 quotes, 20 publishers.

Across 8 chatbots, premium tiers are *more* confidently wrong

Engine	Inaccuracy	Behavior
Perplexity (free)	**37%**	best of eight
ChatGPT Search	67%	hedged 15/200
Perplexity Pro (paid)	higher	decline rate falls
Gemini	high	fabricated URLs frequent
DeepSeek	high	misattributed 115/200
Grok 3 (paid)	**94%**	154/200 broken URLs

Paid tiers answer *more* prompts confidently — so they make *more* confident errors. Robots.txt routinely violated.

Jazwinska & Chandrasekar (2025), *CJR*, Mar. 2025; 1,600 queries (20 publishers × 10 quotes × 8 chatbots).

The economic price falls on the source

Direct answers are paid for in someone else's traffic.

AI summaries cut Wikipedia pageviews ∼15%, costing \$35.7M–\$102.1M per year

Outcome	Estimate	Source
Pageview decline (post-AI summaries)	-15%	DiD, 2024–25
Annual donation revenue at risk	\$35.7M–\$102.1M	extrapolation
Editor-recruitment funnel	weaker	traffic-volunteer link

The first feedback loop: fewer readers → fewer donations and editors → a thinner Wikipedia → a thinner training corpus.

Khosravi et al. (2026); pageview DiD around AI Overviews / Search GPT rollouts.

Sponsored-search auctions splinter: short-query CPC $+1.5\%$, long-query CPC -2.5%

Query length	Δ CPC (BERT)	Mechanism
Short (1–2 tokens)	$+1.5\%$	better intent match $\rightarrow$ tighter rivalry
Long-tail (5+ tokens)	-2.5%	query absorbed by AI answer

The second feedback loop: the highest-intent traffic that publishers monetized via long-tail SEO is exactly what AI answers consume — so its auction price falls.

Larsen et al. (2025); 11,949 queries $\times$ 24 months SEMRush panel; BERT rollout DiD.

Creators adapt — and the well gets polluted

Optimization moves on. The optimization target is now an AI.

Adding a single quotation lifts AI-engine visibility by 41%

GEO tactic	Δ visibility	Cost to publisher
Quotation Addition	+41%	low (one edit)
Statistics Addition	+32%	low
Authoritative tone	+30%	stylistic only
Citing Sources	+27%	low

Generative Engine Optimization (GEO) is now a discipline. The new ranking signal rewards *quotability*, not authority — so creators feed the engine what it likes.

Aggarwal et al. (2024), *KDD '24*; benchmark of 9 GEO tactics on 10K queries.

By late 2024, ~24% of corporate press releases and ~18% of CFPB complaints are LLM-written

Corpus	LLM share	Sample (N)
Corporate press releases	~24%	537,413
Consumer complaints (CFPB)	~18%	687,241
UN press releases	~14%	15,919
Small-firm job postings	~10–15%	1.44M

Distributional MLE; prediction error $< 3.3\%$; *lower bound* (heavy edits invisible). The corpus on which the next models train is increasingly authored *by* models.

Liang et al. (2025), *Patterns*; four corpora, Jan 2019–Sep 2024.

Devil's advocate: aren't most queries the kind where citations don't matter?

The strongest objection

For "what time is the show," "capital of Peru," "recipe for risotto," AI answers users more cheaply than ten blue links — and a wrong citation costs nothing.

Reply. Three problems with the carve-out:

- Liu (2023): $r = -0.96$ between fluency and citation precision — users have *no signal* that flips them from "trust" to "verify."
- Jazwinska (2025): premium-tier engines decline less, so confident wrongness rises with willingness to pay.
- Khosravi (2026) + Larsen (2025): even "trivial" queries strip the long-tail traffic that funds the verifiable corpus.

The cheap query subsidises the expensive one. Sever the cross-subsidy, both fail.

Potthast's prophecy is now a self-reinforcing loop with five turns of the crank

#	Turn	Empirical anchor
1	Citations decay	Liu 2023: precision 74.5%, recall 51.5%, $r = -0.96$
2	Misattribution scales	Jazwinska 24/25: 60%–94% wrong (8 engines)
3	Page views collapse	Khosravi 2026: -15% Wikipedia traffic
4	Ad auctions reprice	Larsen 2025: -2.5% long-tail CPC
5	Synthetic content fills gap	Aggarwal 24 (GEO $+41\%$); Liang 25 (∼24% PR)

The training corpus that produced the model is being eroded by the model. Each turn cheapens the next.

Direct answers are not free.

They are an externality on the web that produced them.

Deep Dive 3

The Closing Web

Consent, Copyright, and the Crisis in AI Training Data

Cooper et al. (2025), *Review of Network Economics* | Yang & Zhang (2024), *arXiv:2402.17801v2*
Fletcher (2024), *Reuters Institute Factsheet* | Kim et al. (2025), *IMC '25* | Longpre et al. (2025), *DPI / MIT*

25.9%

of OpenAI's head-domain training tokens

were withdrawn via robots.txt — in twelve months.

The web is closing faster than any compute curve.

Longpre et al. (2025), Table 3 (HEAD_C4); restriction rate, mid-2023 to April 2024.

AI training rests on four pillars — all four are cracking

Pillar	What it does	Status (this deck)
Consent signal	Tells crawlers what is permitted	Eroding (Longpre 2025)
Voluntary compliance	Bots honor the signal	Selective (Kim 2025)
Premium supply	Provides quality training input	Withdrawing (Fletcher 2024)
Legal backstop	Enforces when norms fail	Fragmenting (Cooper; Yang)

In one year, head-domain training tokens went from 3% restricted to 33%

Corpus slice	Mid-2023	April 2024
C4 / RefinedWeb / Dolma (full)	∼1%	5–7%
HEAD_All (top 2K domains)	<3%	20–33%
SARIMA forecast, April 2025	—	+2–4% (full); +7–11% (head)

Terms of Service now restrict **45–55%** of all corpus tokens — a 26–53% relative jump in one year.

Longpre et al. (2025), Figs. 1–2; pp. 5–7. Audit of C4, RefinedWeb, Dolma.

The closing targets one crawler 26× more than another

Crawler / organization	Head tokens restricted
OpenAI (GPTBot)	25.9%
Anthropic (ClaudeBot)	13.3%
Common Crawl (CCBot)	13.3%
Google-Extended (Google AI)	9.8%
Cohere	4.9%
Meta (FacebookBot)	4.1%
Internet Archive	3.2%
Google Search (Googlebot)	**1.0%**

Longpre et al. (2025), Table 3 (HEAD_C4). Restrictions are AI-specific, not anti-crawl.

But the signal is unenforceable

robots.txt is a polite request, not a wall.

Bot compliance *halves* as robots.txt directives become stricter

Directive	Weighted compliance
Crawl delay (30s)	0.609
Endpoint access restriction	0.310
Disallow all	**0.307**

The protocol fails on contact

The advisory directive (delay) is followed twice as often as the exclusionary one. Switching from "be polite" to "stay out" loses half the leverage.

Kim et al. (2025), Table 5; 3.9M requests, 36 sites, 8 weeks.

Bots' stated policy is uncorrelated with their observed behavior

Bot	Claimed	Observed compliance
ClaudeBot, GPTBot, Amazonbot	respects	**1.000** (disallow-all)
PerplexityBot	refuses	0.897 (endpoint)
Applebot	respects	0.043 (disallow-all)
Bytespider	refuses	0.000 (endpoint)
BrightEdge	respects	0.000 (disallow-all)

Selection effect: legitimate frontier labs are most exposed to opt-outs; lower-reputation actors are not.

Kim et al. (2025), Table 6 & Fig. 9.

Premium supply retreats first

The most valuable content leaves before the average page does.

By end of 2023, 48% of top news sites blocked OpenAI — and none reversed

Outlet type	Blocking OpenAI	Blocking Google AI
Legacy print	57%	32%
Broadcast	48%	19%
Digital-born	31%	17%

- USA leads at 79% OpenAI blocking; Spain, Mexico, Poland trail at 20–27%.
- Across 150 outlets in 10 countries, **no site unblocked** either crawler in 2023.
- Higher-reach outlets block more — *quality* retreats faster than quantity.

Fletcher (2024), Figs. 1, 3, 4. $N = 150$ Wayback-Machine longitudinal sample.

The most-restricted corpus is the corpus AI users barely query

Content category	HEAD_C4 share	ChatGPT query share
News & periodicals	$\sim$40%	<1%
Creative composition	small	$\sim$28%
Sexual roleplay	<1%	12%

The double irony

News publishers are blocking a corpus AI rarely uses at inference; AI usage concentrates in domains for which little permitted training data exists.

Longpre et al. (2025), Fig. 4. WildChat $N = 1$M conversations.

The legal backstop fragments

Three jurisdictions, four rulings, no shared doctrine.

$1.5B already paid for the input stage — and the doctrine is still unsettled

Bartz v. Anthropic *(Alsup)*	Initial book downloads infringing → **$1.5B** (largest in history); training itself ruled fair use.
Kadrey v. Meta *(Chhabria)*	Fair use *only because* "plaintiffs made the wrong arguments"; judge believes training is generally illegal.
USCO *(2025b)*	Doubts fair use applies when outputs compete in markets for training works.
LAION *(Hamburg)*	EU TDM Art. 4: ToS opt-outs likely enforceable.

Cooper et al. (2025), pp. 165–166. Two-thirds of public training datasets carry wrong/missing licenses.

Theory: naive responses backfire

The obvious fix — "just allow more training" — can destroy welfare.

When data is scarce, generous fair use harms AI development

- **Generous fair use** ($f \to 0$): kills creator incentives $\to$ no human content for training $\to$ model improvement stalls. *(Prop. 10)*
- **Strict fair use** (f too large): the AI company abandons model improvement entirely. *(Prop. 11)*
- **A moderate** f maximizes model quality X_2, social welfare, and consumer surplus.

The dynamic problem

A copyright doctrine welfare-optimal in 2020 (data-abundant) becomes welfare-destroying once pre-existing data is exhausted (data-scarce, $Q_0 = 0$) — the regime Longpre et al. document we are entering.

Yang & Zhang (2024), Propositions 10–11; Figure 7.

Strong AI-copyrightability flips fair use from helpful to harmful

AI-copyrightability ϕ	Effect of generous fair use on AI development
Weak	**Promotes** development creators still produce human content for training
Strong	**Hinders** development too few human creators $\rightarrow$ training data dries up

X_2 is inverted-U in ϕ when data is scarce *(Prop. 12)*. The two policy levers cannot be designed independently.

Yang & Zhang (2024), Figure 9 & Corollary 2.

A skeptic would say the closing web is rhetorical, not real

The objection. If half the bots ignore strict directives (Kim), if robots.txt is not legally binding, if frontier labs cut bilateral licenses, then "the web is closing" is a story about polite, compliant labs — not about training in practice.

The response.

- Compliant labs are the regulated mainstream: ClaudeBot, GPTBot and Amazonbot all hit **1.000** on disallow-all (Kim Table 6). Restrictions therefore bite hardest on the firms regulators can actually reach.
- Spoofing is small but selective: <0.1% of traffic is spoofed, yet spoofed bots show "little respect for robots.txt" (Kim Table 9, Fig. 11) — a perverse-incentive layer above the consent system.
- The four pillars are joint constraints: even if signaling and compliance fail, the supply (Fletcher) and legal (Cooper, Yang) pillars still bind. Ignoring robots.txt does not make a $1.5B recovery go away.

Restoring a functioning data commons requires aligning four levers

Lever	Source	Concrete reform
Consent signaling	Longpre; Kim	Successor to REP-1995: machine-readable, granular, use-specific
Compliance	Kim	Legally binding obligations on identified crawlers
Premium supply	Fletcher; Cooper	Collective licensing / structured remuneration
Joint copyright design	Yang; Cooper	Set fair use f and AI-copyrightability ϕ jointly per data regime

Each lever necessary; none sufficient on its own.

The web is not running out of data.

It is running out of consent.

Deep Dive 4

The Productivity Gap

Why GenAI's 80% Exposure Yields a 1.56% Macro Forecast

Eloundou et al. (2024), *Science* 384, 1306 | Noy & Zhang (2023), *Science* 381, 187
Acemoglu (2025), *Economic Policy* 40(121) | Agrawal, Gans & Goldfarb (2025), NBER ETAI
Demirci, Hannane & Zhu (2025), WP | Goldberg & Lam (2026), Stanford GSB / UCLA

80% — 1.56%

Eight in ten US workers have $\geq$10% of their tasks LLM-exposed.
The 10-year GDP gain is bounded at 1.56%.
Both numbers are right.

Eloundou et al. (2024), Fig. 2; Acemoglu (2025), Table 1, p. 51.

Three pieces of math separate exposure from growth

Layer	Question	Number
Exposure	What share of tasks could AI touch?	80% workers, 4.6% of GDP
Lab gain	How fast/better does AI make a task?	-0.8 SD time, $+0.4$ SD grade
Macro envelope	Hulten: $d\,\ln\text{TFP} = \bar{\pi} \times s$	$\leq$ 0.66% TFP / 1.56% GDP
Market reality	Where does the math actually hit?	Narrow sectors, sharp falls
Theory	Who gets displaced first?	Routine, where $\eta_{AI} \leq 1/q^2$

Each row pins down one paper. Read together, they explain why the lab is loud and the macro is quiet.

The headline numbers

Big exposure, big lab gains, small macro envelope.

LLM exposure rises with wages — reversing prior automation's bias

Job zone	Median income	Mean exposure	Share >50%
1 (no prep)	$30,230	0.06	0.0%
2	$38,215	0.16	6.1%
3	$54,815	0.26	10.6%
4 (bachelor's)	**$77,345**	**0.47**	**34.5%**
5 (master's+)	$81,980	0.43	26.5%

34 occupations have zero exposed tasks — all manual: roofers, dishwashers, slaughterers, pile-driver operators. Only 1.86% of all 19,265 tasks are fully automatable (T4).

Eloundou et al. (2024), Tables 1, 5, 8–9; metric is $E1 + 0.5 \cdot E2$.

In the lab, ChatGPT cuts time 0.8 SD and raises grades 0.4 SD

Outcome (Δ post−pre)	**ITT (OLS)**	**TOT (IV)**	N
Time on task (minutes)	-10.84^{***}	-16.51^{***}	346
Grade (1–7 scale)	$+\mathbf{0.65}^{***}$	$+\mathbf{0.95}^{***}$	1,135
Job satisfaction (1–10)	$+1.16^{***}$	$+1.93^{***}$	449

Drafting time halves; editing time doubles. ChatGPT operates as a *drafting substitute*, not an ideation aid — 58 of 173 users submitted output **unedited**.

Noy & Zhang (2023), pre-registered RCT, $N = 450$ Prolific, 5 white-collar tasks; SI Tables S.1–S.2.

Hulten arithmetic bounds 10-year TFP gain at 0.66%

Component	Value	Source
Wage-bill share exposed	20%	Eloundou (2024)
$\times$ cost-effectively automatable	23%	Svanberg et al. (2024)
GDP share affected	$=$ **4.6%**	—
Avg. overall cost savings $\bar{\pi}$	14.4%	Noy/Brynjolfsson
$d \ln$ **TFP** $= s \times \bar{\pi}$	$=$ **0.66%**	p. 41
GDP gain (capital response, 10 yrs)	$\leq$ 1.56%	Table 1, p. 51

An order of magnitude below Goldman's 7% and McKinsey's 1.5–3.4 pp/yr — those forecasts cannot be reconciled with task-level estimates inside Hulten.

Acemoglu (2025), eq. 14 (p. 28); Table 1 cols. 3 & 7 (p. 51).

Where the math hits

The macro is small. The dislocations are not.

ChatGPT cut writing-prone freelance posts 21% within 8 months

Cluster	DiD coef. (SE)	% change	Shock
Writing	-0.362^{***} (0.054)	-30.4%	ChatGPT
Software / web	-0.231^{**} (0.054)	-20.6%	ChatGPT
Engineering	-0.110 (0.058)	-10.4%	ChatGPT
Graphic design	-0.204^{**} (0.048)	-18.5%	Image AI
3D modeling	-0.169^{**} (0.048)	-15.6%	Image AI

Surviving employers post higher budgets (+5.7%), more complex jobs (+2.2%), more bids (+8.6%) — the routine layer is skimmed off, not the complex one.

Demirci et al. (2025), Tables 2–3, D1; $N = 1{,}218{,}463$ posts.

On Adobe Stock, GenAI more than doubled supply and cut non-GenAI artists 29%

Outcome (Post 7+ mth $\times$ Treated)	All images	Non-GenAI
Production (Poisson APE)	$+$**135.9%*****	$-14.7\%^{***}$
Active artists	$+$**47.1%*****	$-29.0\%^{***}$
Sales (Poisson APE)	$+$**82.0%*****	$-28.2\%^{***}$

Expansion and crowd-out coexist. Non-GenAI artists differentiate — spreading in embedding space where GenAI clusters do not yet reach. Embedding quality of remaining non-GenAI rises $+1.8\%^{***}$.

Goldberg & Lam (2026), Tables 2–4; 1,719 markets, $N = 55{,}008$.

Theory closes the gap

Routine is fully displaced; geniuses retreat to the frontier.

Routine workers are fully displaced when AI's curvature $\eta_{AI} \leq 1/q^2$

Regime	Routine	Human genius	AI genius
Pre-AI	Interior $[x_0 - q, x_0 + q]$	By $AA(x)$ peak	—
Short-run AI	Unchanged	Pushed outward	Closer + unanswered
Long-run, $\eta_{AI} \leq 1/q^2$	**Fully displaced**	**Frontier only**	Routine band
Long-run, $\eta_{AI} > 1/q^2$	Narrow edge band	Frontier	Interior

The displacement condition is a geometry inequality, not a productivity threshold: routine's linear value loses wherever AI's quadratic curvature is shallow enough.

Agrawal, Gans & Goldfarb (2025), Propositions 1–3.

A skeptic says: the macro estimate is too low, the J-curve is coming

The objection. The 1.56% ceiling counts only existing tasks at today's ∼14% cost savings. Goldman's 7% and McKinsey's 1.5–3.4 pp/yr could be right if AI generates new tasks the rubric does not see.

The response.

- Adoption is a headwind, not a tailwind. Only 1.5% of US firms had invested in AI by 2019 — 10-year cost reductions are likely *over*stated.
- New tasks have a sign problem. Acemoglu's bad-task calibration: AI-enabled social media + IT-security ≈ 2% of GDP, ratio of harm-to-benefit $-19/53$. Apparent +2% GDP coexists with −0.72% welfare.
- Micro converges with macro. Demirci's 21% drop and Goldberg's 29% artist exit are *narrow* — exactly what Hulten predicts when 4.6% of GDP is exposed at $\bar{\pi} \approx 14\%$.

Bigger gains require new pro-worker tasks — not better chatbots

Layer	What current AI delivers	What would lift the ceiling
Exposure	Broad, progressive (Eloundou)	New-task exposure measures
Lab gain	Drafting substitution (Noy)	Expertise provision: nurses, electricians
Macro	0.66% TFP, 1.56% GDP	New pro-worker tasks Hulten ignores
Market	Routine skimmed (Demirci, Goldberg)	Differentiation niches survive
Theory	Humans $\to$ frontier (Agrawal)	$\eta_{AI} > 1/q^2$: complement, not substitute

The Hulten ceiling is not a law of nature — it is what current AI is built to do.

The macro is small.

The dislocations are not.

Deep Dive 5

Fair Use and AI-Copyrightability Must Be Designed Jointly

Synthesizing Yang & Zhang (2024) and Cooper et al. (2025)

Yang & Zhang (2024), *arXiv:2402.17801v2* | Cooper, Martens, Peukert, & Stocker (2025), *Review of Network Economics* 24(3): 161–176

$1.5 billion

the largest copyright recovery in history

paid by one AI lab — for downloads, not training.

And the doctrine is still unsettled.

Bartz v. Anthropic; Cooper et al. 2025, p. 165

GenAI breaks copyright at both input and output stages

Stage	Core question	Status
Input	Can copyrighted works be scraped for training?	Unsettled: fair use (US), TDM & opt-outs (EU)
Output	Are AI-assisted works copyrightable?	Disparate thresholds: US, EU, China

Cooper et al. (2025), Sections 3.1–3.3.

Two-thirds of AI training datasets carry wrong or missing licenses

- $\sim$**65%** of datasets on Hugging Face & PapersWithCode have missing, incorrect, or ambiguous license metadata.
  (Longpre et al. 2024a, cited in Cooper 2025, p. 167)
- In **LAION-5B** (largest public image set): fewer than **1 in 1,000** datapoints carry license metadata.
- **10 website domains** account for **23%** of LAION's $\sim$2 billion images — supply is concentrated, not representative.
- Opted-out vs. licensed images depict semantically different concepts and differ in technical quality.
  (Shcherbakov et al. 2025, cited in Cooper 2025, pp. 167, 171)

The two leading US rulings disagree on why training is fair use

Alsup (Anthropic)	Initial book downloads infringing → **$1.5B** recovery; subsequent training use ruled fair use.
Chhabria (Meta)	Training fair use *only because* plaintiffs "made the wrong arguments"; judge believes training would generally be illegal.
USCO (2025b)	Doubts fair use applies when AI outputs compete in existing markets for training works.
EU — Hamburg	LAION ruling: terms-of-service opt-outs likely enforceable.

Cooper et al. (2025), pp. 165–166. Dozens of US suits remain pending.

The dynamic problem

What is welfare-maximizing today can be welfare-destroying tomorrow.

When training data is abundant, generous fair use maximizes *every* welfare metric

Welfare variable	**Maximized by**
Model quality X_2	$f = 0 \;\rightarrow\; X_2 = 1$
AI content quantity	$f = 0$
Creator income	$f = 0$
Consumer surplus	$f = 0$ (all tested X_1, k, M)
Social welfare	$f = 0$

Optimal joint policy: $\{f = 0,\ \phi = 1\}$ — generous fair use *plus* full AI-copyrightability.

Yang & Zhang (2024), Proposition 7 & Corollary 2; data-abundant regime, $Q_0 \rightarrow \infty$.

When training data is scarce, generous fair use harms AI development

- **Generous fair use** ($f \to 0$): kills creator incentives $\to$ no human content for training $\to$ model improvement stalls. *(Prop. 10)*
- **Strict fair use** (f too large): AI company abandons model improvement entirely. *(Prop. 11)*
- **A moderate f maximizes** model quality X_2, social welfare, and consumer surplus.

The central novel finding

A copyright doctrine welfare-optimal in 2020 (data-abundant) becomes welfare-destroying once pre-existing data is exhausted (data-scarce, $Q_0 = 0$).

Yang & Zhang (2024), Propositions 10–11.

Strong AI-copyrightability flips fair use from helpful to harmful

AI-copyrightability ϕ	Effect of generous fair use on AI development
Weak	**Promotes** development creators still produce human content for training
Strong	**Hinders** development too few creators $\rightarrow$ training data dries up

X_2 is inverted-U in ϕ when data is scarce *(Prop. 12)*. The two policies cannot be designed independently.

Yang & Zhang (2024), Figure 9 & Proposition 12.

The output side

Three jurisdictions, three verdicts, one work.

Three jurisdictions reach three verdicts on the same AI-assisted work

Jurisdiction	Ruling	Threshold
China *(Li v. Liu, 2023)*	**Copyrightable**	“Intellectual investment” & “personalized expression” via prompting
US *(USCO 2025a)*	**Not copyrightable**	“All current-state prompting outputs” lack sufficient human control
EU *(Czech)*	**Not copyrightable**	“Plaintiff did not personally create the work”

Cooper et al. (2025), Section 3.2, pp. 168–170.

Disclosure cannot enforce the AI / non-AI line — and creates perverse incentives

- **The "Old Hard Drive Debacle"** *(Cooper 2025)*: artists must log every GenAI use solely to strip themselves of rights → strong incentive to *deny* AI use.
- **No robust detection tool** exists for whether — let alone *which elements* of — a work used GenAI.
  (10+ sources on watermarking limits cited in Cooper 2025)
- **Convergence problem:** search-engine and AI-training crawlers are technologically indistinguishable; blocking one blocks the other.

Cooper et al. (2025), Section 3.2, pp. 169–170, 172.

Synthetic data is no escape — model collapse and copy-mining await

- GenAI produces near-identical copies of protected works even with developer safeguards.
  (Cooper & Grimmelmann 2025)
- US "Blurred Lines" precedent: weak similarity thresholds mean outputs need not be close replicas to infringe.
- **Copy-mining incentive**: mass-generate to claim maximum rights, mining out the latent space between existing works.
  (Cooper 2025; Lemley 2024)
- **Synthetic data fails as a fix:** depends on real-data quality, cannot generate genuine novelty, recursive training risks model collapse.
  (Shumailov et al. 2024)

Optimal policy depends on objective *and* data regime — divergence is rational

Regulator's objective	**Optimal ϕ**	**Optimal f**
Consumer surplus	Lowest	More generous
Social welfare / AI development	Substantially higher	Stricter

- Fair use is more sensitive than ϕ to data availability (Q_0), training efficiency (k), and market growth (M).
- Predicted alignment: **US / China** $\rightarrow$ AI growth; **EU** $\rightarrow$ consumer welfare $\rightarrow$ persistent jurisdictional divergence.

Yang & Zhang (2024), Figure 10; Cooper et al. (2025), citing Peukert 2025a.

A skeptic would say the model is too stylized to drive real policy

The objection. Yang & Zhang assume uniform skill, competitive AI markets, binary content quality, and a single regulator. Cooper et al. provide no original empirical evidence.

The response.

- Yang & Zhang's regime-dependence result (abundant vs. scarce) is robust across all numerical examples and does not hinge on the uniform-skill closed form.
- The Cooper primer's job is not estimation — it identifies empirical gaps (representativeness of opt-out data, creator-behavior responses, output memorization rates) for the next generation of work.
- Both papers converge on the same conclusion from independent methods: copyright doctrine cannot be optimized one lever at a time, in one jurisdiction, at one moment in time.

Joint. Dynamic. Cross-border.

Fair use and AI-copyrightability are economically linked.

Designing them in isolation destroys welfare once data runs scarce.

Four crises. One feedback loop.

Trust, displacement, consent, growth — bound by a single policy surface no jurisdiction has yet drawn.